

Worksheet 6 - Solution

PART 1: Mass-Spring Systems

1) In the following system, the input is the angular displacement ϕ of the end of the shaft, and the output is the angular displacement θ of the inertia J . The shafts have torsional stiffness k_1 and k_2 . The equilibrium position corresponds to $\phi = \theta = 0$. Derive the equation of motion and find the transfer function $\Theta(s)/\Phi(s)$.

From the free-body diagram, considering the CW as the positive direction, apply Newton's second law.

$$-k_1(\theta - \phi) - k_2\theta = J\ddot{\theta}$$

The differential equation model of system is obtained.

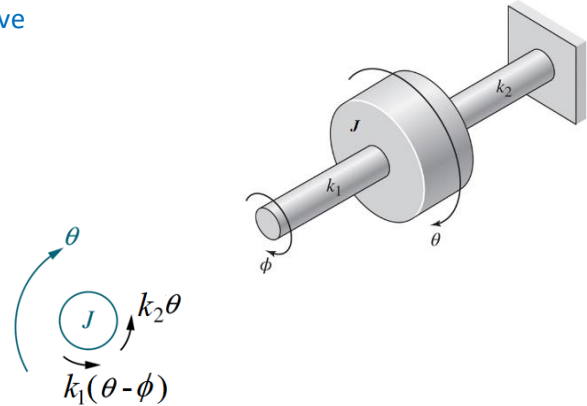
$$J\ddot{\theta} + (k_1 + k_2)\theta = k_1\phi$$

Take Laplace transform:

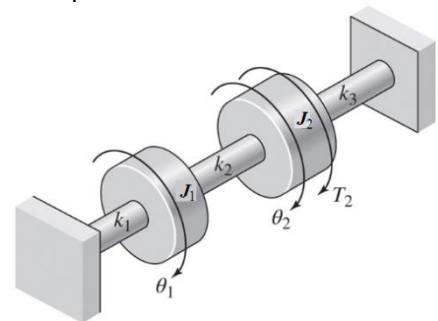
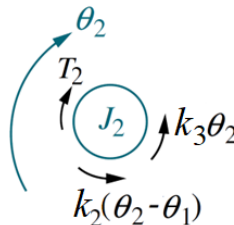
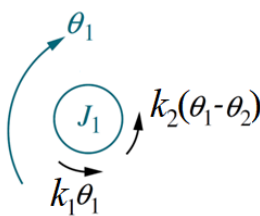
$$Js^2\theta(s) + (k_1 + k_2)\theta(s) = k_1\phi(s)$$

Form the equation as a transfer function model, where the input is the angular displacement ϕ of the end of the shaft, and the output is the angular displacement θ of the inertia J :

$$\frac{\theta(s)}{\phi(s)} = \frac{k_1}{Js^2 + k_1 + k_2}$$



2) The following figure models the three shafts as massless torsional springs. When $\theta_1 = \theta_2 = 0$, the springs are at their free lengths. Derive the equations of motion with the torque τ_2 as the input.



From the free-body diagram, considering the CW as the positive direction, apply Newton's second law:

$$\text{Inertia } J_1 \rightarrow -k_1\theta_1 - k_2(\theta_1 - \theta_2) = J_1\ddot{\theta}_1$$

$$\text{Inertia } J_2 \rightarrow \tau_2(t) - k_2(\theta_2 - \theta_1) - k_3\theta_2 = J_2\ddot{\theta}_2$$

Simplify to obtain the equation of motion of the system

$$J_1\ddot{\theta}_1 + (k_1 + k_2)\theta_1 - k_2\theta_2 = 0$$

$$\tau_2(t) = J_2\ddot{\theta}_2 + (k_2 + k_3)\theta_2 - k_2\theta_1$$

3) Consider the spur gear shown in the following figure. The input shaft (shaft 1) is connected to a motor that produces a torque τ_1 at a speed ω_1 , and drives the output load shaft (shaft 2). One use of such a system is to increase the effective motor torque. θ_1 is the input rotation and θ_2 is the output rotation.

a) Derive the expression for the equivalent inertia J_e felt on the input shaft, and obtain the equation of motion in terms of the displacement θ_1 .

The equivalent inertia felt on the input shaft is:

$$J_e = J_1 + J_2 \left(\frac{N_1}{N_2} \right)^2$$

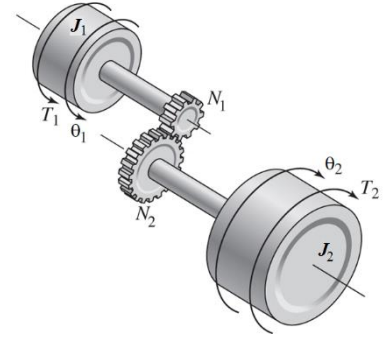
The equivalent torque in the input shaft is:

$$\tau_e = \tau_1 + \tau_2 \left(\frac{N_1}{N_2} \right)$$

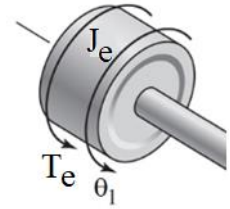
where, $\tau_2 \left(\frac{N_1}{N_2} \right)$ is the reflected torque from shaft 2 to shaft 1.

The equation of motion in terms of θ_1 is:

$$\tau_e = J_e \ddot{\theta}_1 \rightarrow \tau_1 + \tau_2 \left(\frac{N_1}{N_2} \right) = \left[J_1 + J_2 \left(\frac{N_1}{N_2} \right)^2 \right] \ddot{\theta}_1$$



$\frac{\tau_1}{\tau_2} = \frac{\theta_2}{\theta_1} = \frac{N_1}{N_2} = \frac{\omega_2}{\omega_1}$



b) If the inertias are $J_1 = 0.1 \text{ kg.m}^2$ for the motor shaft and $J_2 = 0.4 \text{ kg.m}^2$ for the load shaft. The motor speed ω_1 is five times faster than the load speed ω_2 , so this device is called a *speed reducer*. Obtain the equation of motion in terms of ω_2 . Assuming that the motor torque τ_1 and load torque τ_2 are given.

From the given speed information we have:

$$\frac{\omega_2}{\omega_1} = \frac{1}{5} \rightarrow \frac{N_1}{N_2} = \frac{1}{5}$$

The equivalent inertia is:

$$J_e = J_1 + J_2 \left(\frac{N_1}{N_2} \right)^2 = 0.1 + 0.4 \left(\frac{1}{5} \right)^2 = 0.116 \text{ kg.m}^2$$

Since $\ddot{\theta}_1 = \dot{\omega}_1$, the equation of motion in terms of ω_1 is:

$$\tau_1 + \tau_2 \left(\frac{N_1}{N_2} \right) = J_e \dot{\omega}_1 \rightarrow \tau_1 + \frac{\tau_2}{5} = 0.116 \dot{\omega}_1$$

From the relation of $\omega_1 = 5\omega_2$, we can find the equation of motion in terms of ω_2 :

$$\tau_1 + \frac{\tau_2}{5} = 0.116 \dot{\omega}_1 \rightarrow \tau_1 + \frac{\tau_2}{5} = 0.58 \dot{\omega}_2$$

4) For the system shown in the following figure, assume that the shaft inertias are small. The remaining inertias in $kg.m^2$ are $J_1 = 0.005$, $J_2 = 0.01$, $J_3 = 0.02$, $J_4 = 0.04$, and $J_5 = 0.2$. The speed ratios are:

$$\frac{\omega_1}{\omega_2} = \frac{3}{2}, \quad \frac{\omega_2}{\omega_3} = 2$$

Obtain the equation of motion in terms of ω_3 . Assume that the torque τ is given.

From the speed ratios we have:

$$\frac{N_2}{N_1} = \frac{\omega_1}{\omega_2} = \frac{3}{2} \quad \text{and} \quad \frac{N_3}{N_2} = \frac{\omega_2}{\omega_3} = 2$$

Reflect all inertias to shaft 3.

$$J_e = \left((J_4 + J_1) \left(\frac{N_2}{N_1} \right)^2 + J_2 \right) \left(\frac{N_3}{N_2} \right)^2 + J_3 + J_5$$

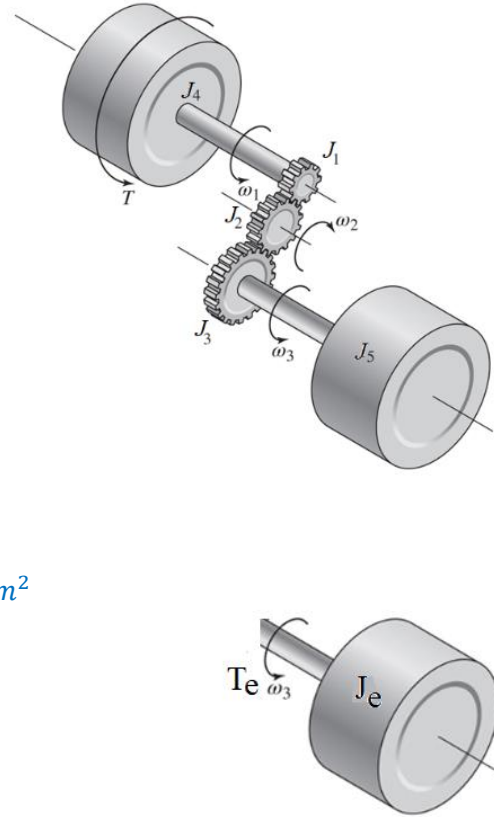
$$J_e = \left((0.04 + 0.005) \left(\frac{3}{2} \right)^2 + 0.01 \right) (2)^2 + 0.02 + 0.2 = 0.665 \, kg.m^2$$

Reflected torque in shaft 3:

$$\tau_e = \tau \left(\frac{N_2}{N_1} \right) \left(\frac{N_3}{N_2} \right) = \tau \left(\frac{3}{2} \right) (2) = 3\tau$$

The equation of motion in terms of ω_3 is:

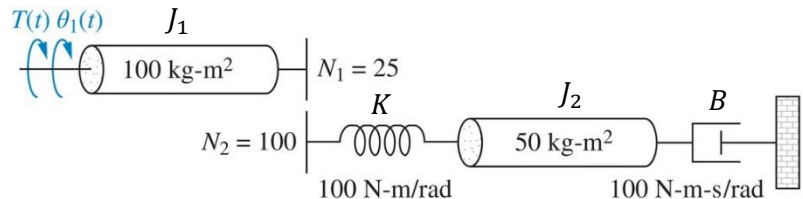
$$\tau_e = J_e \dot{\omega}_3 \quad \rightarrow \quad 3\tau = 0.665 \dot{\omega}_3$$



PART 2: Mass-Spring-Damper Systems

5) Represent the rotational mechanical system shown below in the state-space form, where $\theta_1(t)$ is the output.

Reflect all components and the torque and displacement from input side 1 to side 2:

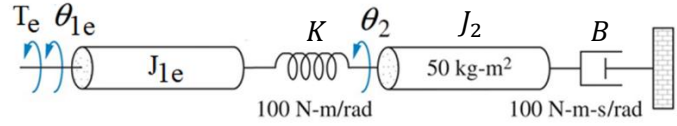


$$\tau_e(t) = \tau(t) \left(\frac{N_2}{N_1} \right) = \tau(t) \left(\frac{100}{25} \right) = 4\tau(t)$$

$$\theta_{1e}(t) = \theta_1(t) \left(\frac{N_1}{N_2} \right) = \theta_1(t) \left(\frac{25}{100} \right) = \frac{1}{4} \theta_1(t)$$

$$J_{1e} = J_1 \left(\frac{N_2}{N_1} \right)^2 = (100) \left(\frac{100}{25} \right)^2 = 1600 \, kg.m^2$$

$$\boxed{\frac{\tau_1}{\tau_2} = \frac{\theta_2}{\theta_1} = \frac{N_1}{N_2} = \frac{\omega_2}{\omega_1}}$$



The equations of motion are:

$$\text{Inertia } J_{1e} \rightarrow \tau_e(t) - K(\theta_{1e}(t) - \theta_2(t)) = J_{1e}\ddot{\theta}_{1e}(t) \rightarrow 4\tau(t) - 100\left(\frac{1}{4}\theta_1(t) - \theta_2(t)\right) = 1600\left(\frac{1}{4}\ddot{\theta}_1(t)\right)$$

$$\text{Eqn. (1)} \rightarrow 400\ddot{\theta}_1(t) + 25\theta_1(t) - 100\theta_2(t) = 4\tau(t)$$

$$\text{Inertia } J_2 \rightarrow -K(\theta_2(t) - \theta_{1e}(t)) - B\dot{\theta}_2(t) = J_2\ddot{\theta}_2(t) \rightarrow -100\left(\theta_2(t) - \frac{1}{4}\theta_1(t)\right) - 100\dot{\theta}_2(t) = 50\ddot{\theta}_2(t)$$

$$\text{Eqn. (2)} \rightarrow 50\ddot{\theta}_2(t) - 25\theta_1(t) + 100\theta_2(t) + 100\dot{\theta}_2(t) = 0$$

Define the state variables:

$$q_1(t) = \theta_1(t)$$

$$q_2(t) = \dot{\theta}_1(t)$$

$$q_3(t) = \theta_2(t)$$

$$q_4(t) = \dot{\theta}_2(t)$$

Find the first derivative of the state variables and rewrite them in terms of the state variables and the input.

$$\dot{q}_1(t) = \dot{\theta}_1(t) \rightarrow \dot{q}_1(t) = q_2(t)$$

$$\dot{q}_2(t) = \ddot{\theta}_1(t) \rightarrow \dot{q}_2(t) = \frac{1}{400}(4\tau(t) - 25\theta_1(t) + 100\theta_2(t)) = 0.01\tau(t) - 0.0625q_1(t) + 0.25q_3(t)$$

$$\dot{q}_3(t) = \dot{\theta}_2(t) \rightarrow \dot{q}_3(t) = q_4(t)$$

$$\dot{q}_4(t) = \ddot{\theta}_2(t) \rightarrow \dot{q}_4(t) = \frac{1}{50}(25\theta_1(t) - 100\theta_2(t) - 100\dot{\theta}_2(t)) = 0.5q_1(t) - 2q_3(t) - 2q_4(t)$$

Find the output in terms of the state variables and the input.

$$y(t) = \theta_1(t) = q_1(t)$$

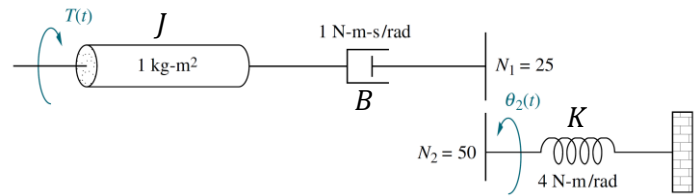
The system model has 4 state variables, 1 input, and 1 output.

Form the state equation and the output equation in the standard matrix-vector form.

$$\text{State Equation} \rightarrow \begin{bmatrix} \dot{q}_1(t) \\ \dot{q}_2(t) \\ \dot{q}_3(t) \\ \dot{q}_4(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -0.0625 & 0 & 0.25 & 0 \\ 0 & 0 & 0 & 1 \\ 0.5 & 0 & -2 & -2 \end{bmatrix} \begin{bmatrix} q_1(t) \\ q_2(t) \\ q_3(t) \\ q_4(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0.01 \\ 0 \\ 0 \end{bmatrix} \tau(t)$$

$$\text{Output Equation} \rightarrow y(t) = [1 \ 0 \ 0 \ 0] \begin{bmatrix} q_1(t) \\ q_2(t) \\ q_3(t) \\ q_4(t) \end{bmatrix} + [0]\tau(t)$$

6) Find the transfer function $\theta_2(s)/T(s)$, for the rotational mechanical system with gears as shown below:

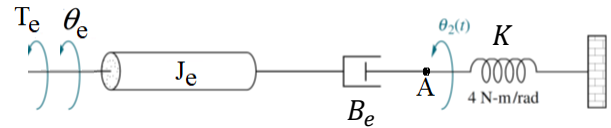


Reflect all components and the torque from side 1 to side 2:

$$\tau_e(t) = \tau(t) \left(\frac{N_2}{N_1} \right) = \tau(t) \left(\frac{50}{25} \right) = 2\tau(t)$$

$$J_e = J \left(\frac{N_2}{N_1} \right)^2 = (1) \left(\frac{50}{25} \right)^2 = 4 \text{ kg.m}^2$$

$$B_e = B \left(\frac{N_2}{N_1} \right)^2 = (1) \left(\frac{50}{25} \right)^2 = 4 \text{ N.m.s/rad}$$



Find the equations of motion:

$$\text{Inertia } J_e \rightarrow \tau_e(t) - B_e (\dot{\theta}_e(t) - \dot{\theta}_2(t)) = J_e \ddot{\theta}_e(t) \rightarrow 2\tau(t) - 4 (\dot{\theta}_e(t) - \dot{\theta}_2(t)) = 4\ddot{\theta}_e(t)$$

$$\text{Point A} \rightarrow -B_e (\dot{\theta}_2(t) - \dot{\theta}_e(t)) - K(\theta_2(t)) = 0 \rightarrow 4 (\dot{\theta}_2(t) - \dot{\theta}_e(t)) + 4\theta_2(t) = 0$$

Taking Laplace transform and form as a transfer function model:

$$2T(s) - 4s\theta_e(s) + 4s\theta_2(s) = 4s^2\theta_e(s) \quad \text{Eqn. (1)}$$

$$4s\theta_2(s) - 4s\theta_e(s) + 4\theta_2(s) = 0 \quad \text{Eqn. (2)}$$

Find $\theta_e(s)$ from Eqn. (2) and substitute in Eqn. (1):

$$\text{From Eqn. (2)} \rightarrow \theta_e(s) = \frac{(s+1)}{s} \theta_2(s)$$

$$2T(s) - 4s \left(\frac{(s+1)}{s} \theta_2(s) \right) + 4s\theta_2(s) = 4s^2 \left(\frac{(s+1)}{s} \theta_2(s) \right)$$

$$\rightarrow 2T(s) - 4(s+1)\theta_2(s) + 4s\theta_2(s) = 4s(s+1)\theta_2(s)$$

$$\rightarrow 2T(s) - 4s\theta_2(s) - 4\theta_2(s) + 4s\theta_2(s) = 4s^2\theta_2(s) + 4s\theta_2(s)$$

$$\rightarrow \frac{\theta_2(s)}{T(s)} = \frac{2}{4s^2 + 4s + 4} = \frac{1/2}{s^2 + s + 1}$$

7) The following figure shows a drive train with a spur-gear pair. The first shaft turns N_2/N_1 times faster than the second shaft. Develop a model of the system including the elasticity of the second shaft. Assume the first shaft is rigid, and neglect the gear and shaft masses. The input is the applied torque τ_1 . The outputs are the angles θ_1 and θ_3 .

Assume that $\frac{N_2}{N_1} = N$

Reflect all components and the torque and the displacement from side 1 to side 2:

$$\tau_{1e}(t) = \tau_1(t) \left(\frac{N_2}{N_1} \right) = N\tau_1(t)$$

$$\theta_2(t) = \theta_1(t) \left(\frac{N_1}{N_2} \right) = \frac{1}{N} \theta_1(t)$$

$$J_{1e} = J_1 \left(\frac{N_2}{N_1} \right)^2 = N^2 J_1$$

Find the equation of motion:

$$\begin{aligned} \text{Inertia } J_{1e} \rightarrow \tau_{1e}(t) - k_T(\theta_2(t) - \theta_3(t)) &= J_{1e}\ddot{\theta}_2(t) \\ \rightarrow N\tau_1(t) - k_T\left(\frac{1}{N}\theta_1(t) - \theta_3(t)\right) &= (N^2 J_1) \left(\frac{1}{N}\ddot{\theta}_1(t)\right) \\ \rightarrow N^2\tau_1(t) - k_T(\theta_1(t) - N\theta_3(t)) &= N^2 J_1 \ddot{\theta}_1(t) \quad \text{Eqn. (1)} \end{aligned}$$

$$\begin{aligned} \text{Inertia } J_2 \rightarrow -k_T(\theta_3(t) - \theta_2(t)) - c_T(\dot{\theta}_3(t)) &= J_2\ddot{\theta}_3(t) \\ \rightarrow -k_T\left(\theta_3(t) - \frac{1}{N}\theta_1(t)\right) - c_T(\dot{\theta}_3(t)) &= J_2\ddot{\theta}_3(t) \\ \rightarrow -k_T\left(\theta_3(t) - \frac{1}{N}\theta_1(t)\right) - c_T(\dot{\theta}_3(t)) &= J_2\ddot{\theta}_3(t) \quad \text{Eqn. (2)} \end{aligned}$$

The system model consists of Eqn. (1) and Eqn. (2).

$$\begin{cases} N^2\tau_1(t) - k_T(\theta_1(t) - N\theta_3(t)) = N^2 J_1 \ddot{\theta}_1(t) \\ -k_T\left(\theta_3(t) - \frac{1}{N}\theta_1(t)\right) - c_T(\dot{\theta}_3(t)) = J_2\ddot{\theta}_3(t) \end{cases} \rightarrow \begin{cases} N^2\tau_1(t) = N^2 J_1 \ddot{\theta}_1(t) - N\theta_3(t) + k_T\theta_1(t) \\ J_2\ddot{\theta}_3(t) + c_T\dot{\theta}_3(t) + k_T\theta_3(t) - \frac{k_T}{N}\theta_1(t) = 0 \end{cases}$$

